

What Makes a Song Popular?

A Data Analysis of Spotify Tracks

DSA210 — Introduction to Data Science
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GitHub: <https://github.com/Stirpe999/AkiSh98s-DSA210-repo>

1. Motivation

I have always been curious about what makes some songs become really popular while others with similar sounds stay under the radar. With millions of songs available on Spotify, it seems like there should be some patterns in the audio features that separate popular songs from the rest.

The main goal of this project is to find out whether audio features like energy, danceability, loudness, and acousticness are actually different between popular and unpopular songs, and if so, whether we can use these features to predict popularity. I think this is an interesting problem because it connects music, which is something personal and cultural, with data that can actually be measured.

From a data science perspective, this project covers the full pipeline from raw data to machine learning, which made it a good fit for the course goals.

2. Data Source

The dataset I used comes from Kaggle and was originally created using the Spotify Web API:

Dataset: Spotify Music Dataset by Solomon Ameh

Link: <https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset>

The dataset comes as two separate CSV files:

high_popularity_spotify_data.csv — songs with high popularity scores

low_popularity_spotify_data.csv — songs with low popularity scores

Each song has a set of audio features that Spotify calculates automatically through its audio analysis system. These features describe the sonic characteristics of each track in a numerical way. The table below shows the main features available:

Feature	Description
danceability	How suitable a track is for dancing (0.0 to 1.0)
energy	Intensity and activity level of the track (0.0 to 1.0)
loudness	Overall loudness in decibels (dB)
tempo	Estimated beats per minute (BPM)
valence	Musical positiveness (0.0 = sad, 1.0 = happy)
acousticness	Confidence the track is acoustic (0.0 to 1.0)
instrumentalness	Predicts whether the track has no vocals (0.0 to 1.0)
liveness	Detects presence of a live audience (0.0 to 1.0)
speechiness	Detects spoken words in a track (0.0 to 1.0)
duration_ms	Duration of the track in milliseconds
playlist_genre	Genre of the playlist the track belongs to

I uploaded the CSV files directly into Google Colab using the built-in file upload feature. No scraping or extra data collection was needed since the dataset was already clean and structured.

3. Data Analysis

The analysis was done in four steps, each in a separate Python notebook.

3.1 Data Collection and Initial Checks

After loading the two CSV files, I checked the basic structure of the data:

- Verified the shape (number of rows and columns) for both files
- Looked at the column names to understand what features are available
- Checked for missing values in all columns

There were no major issues. Both files had consistent columns and very few missing values, so I could move on to EDA without much cleaning.

3.2 Exploratory Data Analysis (EDA)

I focused the EDA on five features that seemed most relevant based on the project proposal: energy, danceability, tempo, loudness, and valence. The goal was to understand whether these features look different between the two groups before doing any formal tests.

The steps I followed:

- Computed summary statistics (mean, standard deviation, min, max) for each group separately
- Compared the mean values between high and low popularity songs for each feature
- Plotted overlapping histograms to compare distributions visually
- Made bar charts to compare the group means side by side
- Computed correlation matrices to see how features relate to each other within each group

The EDA already showed some clear patterns. High popularity songs were generally louder and more energetic, while low popularity songs tended to be more acoustic and more instrumental. These early observations motivated the hypothesis tests in the next step.

3.3 Hypothesis Testing

I defined three main hypotheses based on what I observed in the EDA:

Hypothesis	Features Tested	Expected Direction
H1	Energy, Loudness	Higher in high-popularity songs
H2	Danceability, Valence	Higher in high-popularity songs
H3	Acousticness, Instrumentalness	Lower in high-popularity songs

Hypothesis	Features Tested	Expected Direction
H0 (Null)	All features above	No significant difference between groups

I used Welch's t-test instead of the standard t-test because the two groups have different sample sizes and the variances are not necessarily equal. Welch's version handles this better.

For each feature, I also calculated:

- The mean difference between the two groups
- A 95% confidence interval around that difference
- Cohen's d to measure effect size (how big the difference actually is)
- Bonferroni-corrected p-values to account for testing multiple features at once

All three hypotheses were supported. Energy, loudness, acousticness, and instrumentality showed the strongest differences with large effect sizes. Danceability and valence also had statistically significant differences, though the effect sizes were smaller.

3.4 Machine Learning

I applied machine learning in two ways: classification (predict whether a song is high or low popularity) and regression (predict the actual popularity score).

Classification: The target variable was the binary group label (1 = high, 0 = low). I used an 80/20 train-test split, stratified by class. Four models were tested: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.

Regression: The target variable was the numeric popularity score. I tested Linear Regression, Ridge, Lasso, ElasticNet, Random Forest Regressor, Gradient Boosting Regressor, and SVR.

All models used a preprocessing pipeline with median imputation and standard scaling for numeric features, and one-hot encoding for the genre column. This made the code cleaner and prevented data leakage.

I also visualized the results with confusion matrices, ROC curves, feature importance charts, actual vs. predicted scatter plots, and residual histograms.

4. Findings

4.1 What the EDA Showed

Feature	High Popularity	Low Popularity

Feature	High Popularity	Low Popularity
Energy	Higher	Lower
Loudness	Higher (less negative dB)	Lower
Danceability	Higher	Lower
Valence	Higher	Lower
Acousticness	Lower	Higher
Instrumentalness	Lower	Higher

The direction of these differences was consistent and made intuitive sense. Popular songs on Spotify tend to be louder, more energetic, and more danceable. Less popular songs tend to be quieter, more acoustic, and more instrumental (which could include classical, jazz, or ambient)

4.2 Hypothesis Test Results

H1 supported: Energy and loudness were significantly higher in the high-popularity group ($p \ll 0.05$, large Cohen's d).

H2 supported: Danceability and valence were also significantly higher ($p < 0.05$), though the effect was smaller.

H3 supported: Acousticness and instrumentalness were significantly lower in the high-popularity group ($p \ll 0.05$, large Cohen's d).

The null hypothesis was rejected for all six features, even after applying Bonferroni correction. This means the differences between groups are unlikely to be due to random chance.

4.3 Machine Learning Results

For classification, ensemble methods (Random Forest and Gradient Boosting) performed best in terms of accuracy and F1-score. Logistic Regression also worked reasonably well, showing that a linear model can already capture a meaningful part of the signal.

Feature importance analysis showed that energy, loudness, and acousticness were the most influential predictors, which aligns well with the hypothesis testing results.

For regression, predicting the exact numeric popularity score was harder. The R^2 values were lower than in classification, which makes sense since even within each group the popularity scores vary quite a bit. Ensemble models again outperformed linear ones.

5. Limitations and Future Work

5.1 Limitations

Only two groups: The dataset splits songs into just high or low popularity, so there is no nuance for songs that fall in the middle range.

No context features: Things like artist fame, release date, and marketing effort all affect popularity a lot, but none of that information is in this dataset.

Static snapshot: Popularity changes over time. The dataset does not capture how a song's popularity evolves after release.

Correlation, not causation: Even though the analysis found statistically significant differences, that does not mean that making a song louder will make it popular. Both could be driven by the same underlying trends.

5.2 Future Work

- Add artist-level features like follower count or number of previous hits.

- Include release year to see how the features of popular songs have changed over time.

- Try using lyrics or raw audio data as additional input features.

- Compare results across different genres separately, since what makes a popular pop song might be different from what makes a popular jazz song.

6. Usage of AI Tools

I used AI tools (specifically LLMs) in a few places during this project. Mainly I used them to help debug some Python errors in the notebooks and to double-check my understanding of certain statistical concepts like Welch's t-test and Cohen's d.

All the analysis decisions, the hypothesis definitions, the model choices, and the interpretations of results are my own work.

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